INTEGRATED PLATFORM FOR AI-ASSISTED ANIMATION PRODUCTION AND INTELLECTUAL PROPERTY MANAGEMENT
FIELD OF THE INVENTION

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ABSTRACT

ABSTRACT OF THE DISCLOSURE

An integrated platform for AI-assisted animation production and intellectual property management is disclosed. The system orchestrates an end-to-end animation pipeline, utilizing artificial intelligence for script generation, storyboard creation, voice synthesis, and lip-sync video generation. A character consistency management module ensures visual coherence across generated content using canonical descriptions. The platform features a hierarchical cost modeling system that simulates asset decay and learning curves to optimize production budgets. Furthermore, an integrated intellectual property suite facilitates bulk copyright registration with AI authorship disclosure compliance and automated patent generation via codebase analysis. The system provides a unified environment for democratizing high-quality animation production while safeguarding creative and technical assets.

TITLE OF THE INVENTION

INTEGRATED PLATFORM FOR AI-ASSISTED ANIMATION PRODUCTION AND INTELLECTUAL PROPERTY MANAGEMENT

FIELD OF THE INVENTION

The present invention relates generally to computer-implemented systems and methods for digital content creation and intellectual property management. More particularly, the invention pertains to an integrated platform for orchestrating AI-assisted animation production pipelines, optimizing production costs through hierarchical modeling, and automating the protection of intellectual property rights for generated assets.

BACKGROUND OF THE INVENTION

The creation of animated content is a complex and resource-intensive endeavor, residing at the intersection of artistic expression and sophisticated technical execution. This field encompasses a broad spectrum of techniques, from traditional hand-drawn animation to advanced computer-generated imagery (CGI), all aimed at producing dynamic visual narratives. Concurrently, the rapid evolution of artificial intelligence (AI) and machine learning technologies has begun to influence various aspects of content creation, offering new paradigms for automation and efficiency. Parallel to these developments, the management and protection of intellectual property (IP) in the digital age, particularly for creative works and underlying technologies, has become increasingly critical, necessitating robust systems for copyright, trademark, and patent

Conventionally, animation production workflows are characterized by a highly segmented and labor-intensive process. A typical pipeline involves numerous specialized roles, including scriptwriters, storyboard artists, character designers, animators, voice actors, sound engineers, and post-production specialists. Each stage often requires significant manual effort and iterative revisions. For instance, script development is typically followed by manual storyboarding to visualize scenes, character design and rigging, keyframe animation, and meticulous lip-synchronization of character movements with recorded dialogue. Existing animation software provides powerful tools for artists to execute these tasks, but these are primarily creation environments rather than integrated automation platforms. While some systems, such as those described in U.S. Pat. No. 10,783,691, address aspects like maintaining visual consistency in computer-generated imagery, they typically operate within traditional rendering pipelines rather than leveraging advanced AI for prompt-based generation and cloud-based asset management.

Furthermore, the landscape of content creation has seen the emergence of AI-powered tools for generating isolated media elements. For example, text-to-speech services can synthesize voices, and certain AI models can generate short video clips or images from textual prompts. Systems for automated content production and cost estimation, such as that outlined in U.S. Pat. No. 11,348,209, exist to provide financial oversight. Similarly, AI-powered script analysis and scene breakdown systems (e.g., U.S. Pat. No. 11,657,231) assist in production planning. However, existing systems fail to adequately address the holistic challenges of modern animation production. The traditional animation pipeline remains prohibitively expensive, with costs frequently ranging from $50,000 to $500,000 per minute of finished content. Moreover, contemporary AI video generation services are limited by their inability to maintain consistent character appearances, voices, and narrative continuity across multiple scenes or entire episodes.

The limitations extend significantly into the domain of intellectual property protection. Content creators utilizing AI-assisted tools face an increasingly complex legal environment, particularly concerning copyright ownership and attribution. Current copyright registration systems are largely disconnected from content production platforms, necessitating manual data entry of work details. There is a notable absence of systems offering bulk copyright registration by automatically scanning asset libraries, or integrated trademark application generation with appropriate classification for character names. Furthermore, for software development teams, the process of patenting their own inventions remains a costly endeavor. Existing automated patent drafting systems (e.g., U.S. Pat. No. 11,915,708) typically lack the capability to automatically extract technical details directly from a codebase to provide comprehensive novelty scoring.

Consequently, there is a pressing need for a transformative solution that can drastically reduce the cost and time associated with professional animation production while providing a unified, intelligent environment for orchestrating the entire workflow and securing intellectual property rights.

SUMMARY OF THE INVENTION

The present invention provides an advanced artificial intelligence (AI)-powered platform designed to comprehensively automate and streamline the entire animation production pipeline, from initial script conception to final video generation. This innovative system significantly reduces traditional animation production costs while maintaining professional quality through sophisticated AI assistance combined with strategic human oversight. Furthermore, it integrates a robust intellectual property (IP) protection suite, enabling creators to efficiently produce, manage, and protect their animated content, characters, and scripts within a unified environment.

In one embodiment, a multi-stage AI pipeline orchestrates the end-to-end animation workflow. This pipeline commences with AI-Powered Script Generation, followed by Automated Storyboard Creation, Character Voice Synthesis, and Lip Sync Video Generation. Throughout the production process, the system incorporates character consistency management utilizing canonical descriptions and reference image URIs.

A distinguishing feature of the invention is its sophisticated Hierarchical Cost Modeling system, which incorporates a novel concept of "Asset Decay," simulating the depreciation and reuse value of creative assets over time, alongside learning curve simulations for production processes.

Beyond production, the platform integrates a comprehensive IP Protection Suite. This suite provides integrated tools for managing Patents, Copyrights, and Trademarks. It features a Bulk Copyright Registration system that automatically scans the platform's asset library and auto-populates registration fields, explicitly handling AI Authorship Disclosure Compliance. Additionally, the suite features an Automated Patent Generation System that leverages self-documenting codebase analysis to extract technical details and functional descriptions directly from the platform's architecture.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

FIG. 1 is a high-level block diagram illustrating the overall system architecture of the AI-assisted animation production and IP management platform.

FIG. 2 is a flowchart illustrating the multi-stage AI pipeline for end-to-end animation workflow, from script generation to video synthesis.

FIG. 3 is a schematic diagram of the Hierarchical Cost Modeling system, detailing the inputs and outputs for asset decay and production cost comparison.

FIG. 4 is a block diagram of the Character Consistency Management module.

FIG. 5 is a diagram illustrating the Prompt Template Versioning system and deployment mechanism.

FIG. 6 is a user interface representation of the Real-Time Episode Progress Tracking module.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides a comprehensive system and method for AI-assisted animation production with integrated intellectual property portfolio management.

I. Animation Production Pipeline

At the core of the system is a multi-stage AI pipeline that orchestrates the end-to-end animation workflow. As illustrated in FIG. 2, this pipeline commences with AI-Powered Script Generation, which leverages advanced large language models (LLMs), such as Google Gemini 2.0 Flash or similar transformer-based architectures, to rapidly develop narrative content based on user prompts.

Following script generation, an Automated Storyboard Creation module visualizes these scripts, translating narrative elements into visual sequences. This module utilizes natural language processing (NLP) to parse the script and text-to-image generation algorithms to create panels determining optimal camera angles and character poses.

Character Voice Synthesis is then performed using integrated services (e.g., ElevenLabs Text-to-Speech, Chatterbox TTS) to generate high-quality voiceovers. The final stage involves Lip Sync Video Generation, where AI models (e.g., Google Veo 3, SyncLabs) combine synthesized voices with character animations. This process includes multi-provider lip sync orchestration with unified status tracking.

To ensure visual and narrative coherence, as shown in FIG. 4, the system includes a Character Consistency Management module. This module utilizes "canonical descriptions" and reference image URIs to maintain consistent character appearances and attributes. When generating new scenes, the AI models consult these descriptions to prevent "drift" in character appearance.

II. Hierarchical Cost Modeling and Optimization

A distinguishing feature of the invention, illustrated in FIG. 3, is its sophisticated Hierarchical Cost Modeling system. This module incorporates a novel concept of Asset Decay, which simulates the depreciation and reuse value of creative assets over time. The Asset Decay algorithm calculates an asset's current value based on its initial cost, usage history, and a configurable decay rate (linear, exponential, or logarithmic). Mathematically, the current value of an asset at time t can be expressed as:

$$V(t) = \max(Floor_Value, Initial_Cost \times (1" Decay_Rate)^t)$$

This allows the system to accurately model the "cost" of reusing an asset versus creating a new one. Complementing this, the system includes learning curve simulations, where the cost per unit decreases as cumulative production doubles, following a power law relationship.

III. Intellectual Property Protection Suite

The platform integrates a comprehensive IP Protection Suite designed to automate legal workflows.

A. Bulk Copyright Registration The system features a Bulk Copyright Registration module that automatically scans the platform's asset library—comprising generated scripts, character models, and animation sequences. It employs metadata extraction algorithms to auto-populate copyright registration fields. Crucially, the system incorporates an AI Authorship Disclosure Compliance algorithm. This algorithm tracks and documents the extent of AI involvement (e.g., percentage of generated tokens, AI-driven visual iterations) to ensure compliance with USPTO and Copyright Office requirements regarding human authorship.

B. Automated Patent Generation The suite includes an Automated Patent Generation System. This system leverages self-documenting codebase analysis to extract technical details and functional descriptions directly from the platform's architecture. It integrates a real-time Prior Art Search and Analysis engine with relevance scoring, and a Novelty Scoring Algorithm that compares extracted features against identified prior art. The system utilizes Natural Language Generation (NLG) to draft patent claims and technical specifications.

IV. System Architecture and Management

As shown in FIG. 1, the overall architecture comprises the AI production core, the IP management module, and the cost modeling system. The platform features Prompt Template Versioning (see FIG. 5) with atomic deployment switching, allowing for controlled updates to AI prompts without disrupting active production.

FIG. 6 illustrates the Real-Time Episode Progress Tracking module, which aggregates status across distributed job queues. This provides up-to-the-minute insights into the production status of each episode, utilizing batch processing optimization to schedule computational resources efficiently.

ABSTRACT OF THE DISCLOSURE

An integrated platform for AI-assisted animation production and intellectual property management is disclosed. The system orchestrates an end-to-end animation pipeline, utilizing artificial intelligence for script generation, storyboard creation, voice synthesis, and lip-sync video generation. A character consistency management module ensures visual coherence across generated content using canonical descriptions. The platform features a hierarchical cost modeling system that simulates asset decay and learning curves to optimize production budgets. Furthermore, an integrated intellectual property suite facilitates bulk copyright registration with AI authorship disclosure compliance and automated patent generation via codebase analysis. The system provides a unified

CLAIMS

1. A computer-implemented method for automated animation production with cost optimization, comprising: receiving script input comprising dialogue and scene descriptions; parsing the script into structured data; generating image prompts using character consistency profiles with reference images; transmitting prompts to AI image generation services; synthesizing character dialogue using voice profiles; generating lip-synchronized videos; calculating production costs using hierarchical asset decay modeling; and assembling assets into completed episodes with progress tracking.

2. A system for AI-assisted animation production, comprising: one or more processors; a non-transitory computer-readable storage medium storing instructions that cause the system to: maintain character consistency database with reference images and prompt templates; implement multi-version prompt management with atomic deployment; orchestrate multiple AI service providers through unified abstraction layer; apply cost calculation engine with configurable decay rates and floor values; track episode production progress across multiple job types; and coordinate assembly of generated assets into completed animated content.

3. The method of claim 1, wherein the hierarchical asset decay modeling comprises: calculating decay multiplier as max(floor_value, decay_rate^(episode_number - 1)); applying decay to human editing costs; maintaining flat costs for supervision; and generating comparative cost analysis.

4. The method of claim 1, wherein maintaining character consistency comprises: storing reference image URLs in cloud storage; generating prompts incorporating reference URLs; tracking consistency scores; and updating profiles based on usage patterns.

5. The method of claim 1, wherein the prompt resolution engine implements a hierarchical lookup strategy comprising: checking an in-memory cache; querying an organization-specific prompt table upon cache miss; retrieving a system default prompt upon absence; and caching the retrieved prompt with a configurable time-to-live value.

6. The method of claim 1, wherein generating image prompts comprises: constructing prompts that include character reference image URLs; incorporating physical description attributes from a character database; applying series-specific style guides; and transmitting the prompts to external AI image generation services.

7. The method of claim 1, wherein calculating production costs comprises: implementing an asset decay model with exponential decay rate; tracking API costs per generation request; monitoring human editing time requirements; and generating cost projections over multiple episodes.

8. The method of claim 1, wherein orchestrating external AI services comprises: maintaining a unified abstraction layer; implementing automatic failover logic; applying rate limiting with token bucket algorithm; and tracking usage metrics per service provider.

9. The method of claim 1, wherein synthesizing audio files comprises: mapping characters to voice provider identifiers; supporting multiple voice synthesis providers; enabling voice cloning from sample audio; and generating lip synchronization timing data.

10. The method of claim 1, further comprising: validating total runtime against target duration; identifying runtime discrepancies; generating recommendations for dialogue adjustment; and tracking episode completion progress across multiple generation stages.

11. The method of claim 1, wherein parsing the script input comprises: tokenizing text using regular expressions; identifying act boundaries and scene transitions; extracting character dialogue with speaker attribution; and structuring visual descriptions for image generation.

12. The system of claim 2, wherein the multi-version prompt management comprises: storing multiple versions per prompt template; marking one version as deployed; enabling atomic deployment switching; and supporting organization-level overrides.

13. The system of claim 2, wherein the unified abstraction layer comprises: provider interface defining common operations; provider-specific adapters; automatic failover on service errors; and cost-based provider selection.

14. The system of claim 2, wherein the character consistency database comprises: a relational database storing character profiles; reference image URLs with cloud storage integration; physical attribute schemas; and version tracking for character design iterations.

15. The system of claim 2, wherein the cost calculation engine comprises: a decay rate configuration interface; episode-level cost tracking tables; comparative analysis algorithms; and export functionality for financial reporting.

16. The system of claim 2, wherein coordinating assembly comprises: a job queue system managing generation tasks; progress tracking with percentage completion; error handling with retry logic; and final video compilation with timeline synchronization.

17. The system of claim 2, further comprising: a caching layer reducing redundant API calls by at least 40%; a backup and recovery system for production data; and a multi-tenant architecture supporting organization isolation.

FIG. 1 - High-Level System Architecture Overview

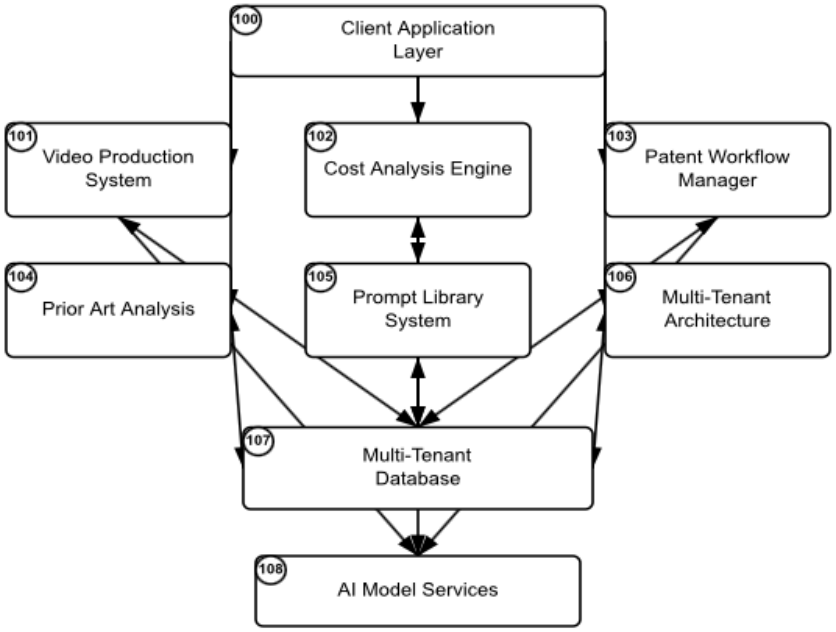

FIG. 1a

FIG. 1a is a block diagram illustrating the high-level system architecture overview showing major subsystems of the integrated video production and patent management platform according to an embodiment of the present invention.

FIG. 2 - Hierarchical Asset Decay System

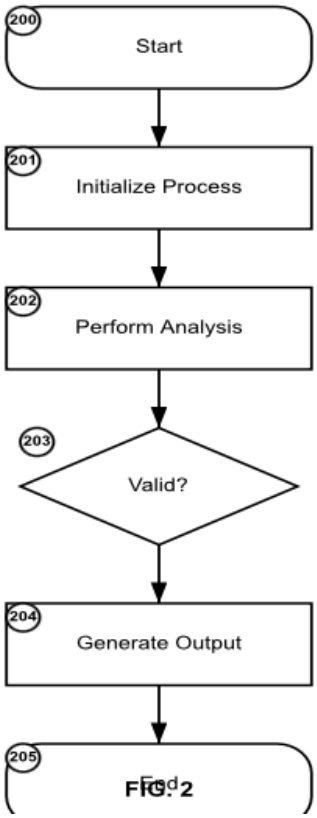

FIG. 2 is a flowchart illustrating the hierarchical asset decay system according to an embodiment of the present invention.

FIG. 3 - Multi-Tiered Cost Comparison Engine

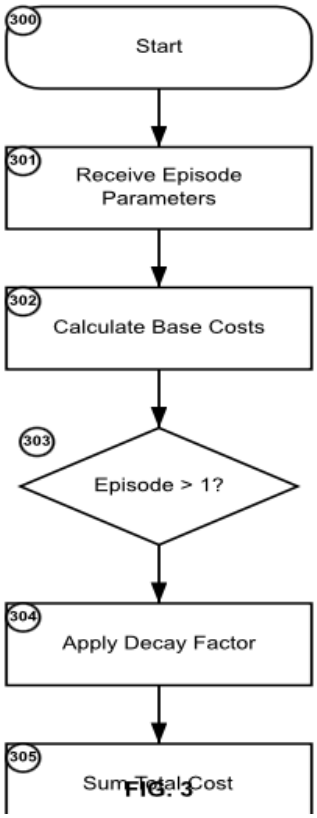

FIG. 3 is a flowchart illustrating the multi-tiered cost comparison engine according to an embodiment of the present invention.

FIG. 4 - Character Consistency Profile System

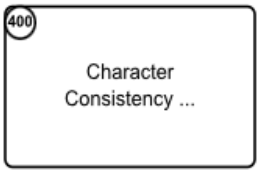

FIG. 4

FIG. 4 is a block diagram illustrating the character consistency profile system according to an embodiment of the present invention.

FIG. 5 - Script-to-Shot Extraction with Dialogue Mapping

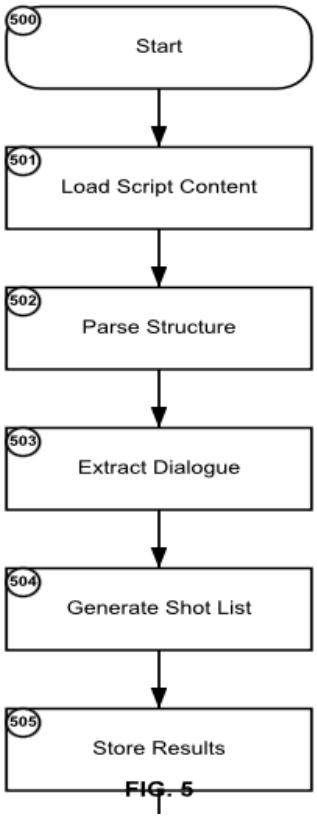

FIG. 5 is a flowchart illustrating the script-to-shot extraction with dialogue mapping according to an embodiment of the present invention.

FIG. 6 - AI-Powered Patent Specification Generation System

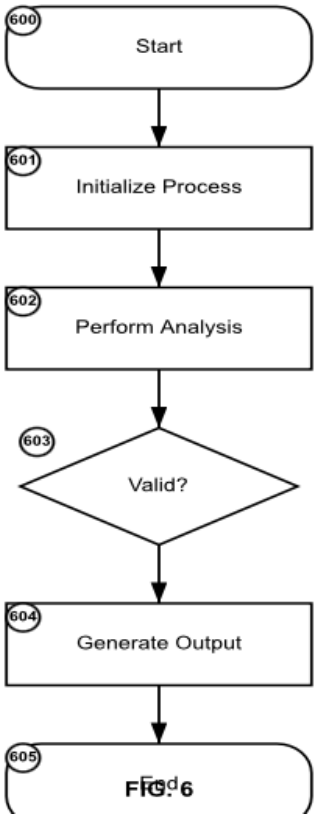

FIG. 6 is a flowchart illustrating the ai-powered patent specification generation system according to an embodiment of the present invention.

FIG. 7 - Real-Time Prior Art Search and Analysis Engine

FIG. 7

FIG. 7 is a block diagram illustrating the real-time prior art search and analysis engine according to an embodiment of the present invention.

FIG. 8 - Feature-Based Novelty Scoring Algorithm

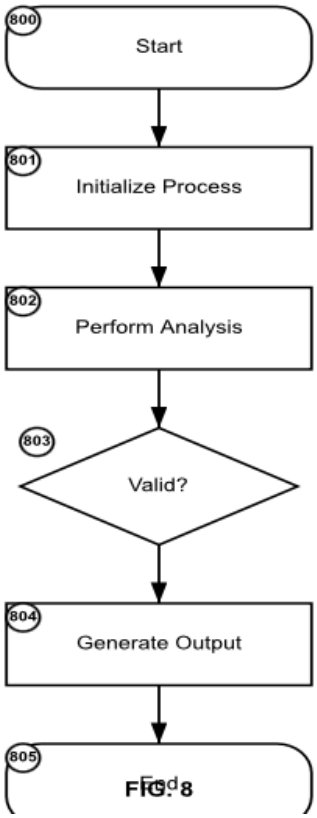

FIG. 8 is a flowchart illustrating the feature-based novelty scoring algorithm according to an embodiment of the present invention.

FIG. 9 - Codebase Feature Extraction with Technical Mapping

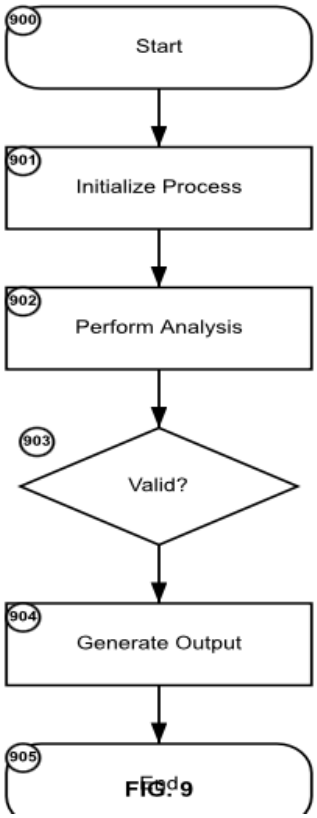

FIG. 9 is a flowchart illustrating the codebase feature extraction with technical mapping according to an embodiment of the present invention.

FIG. 10 - Intelligent Patent Claims Generation

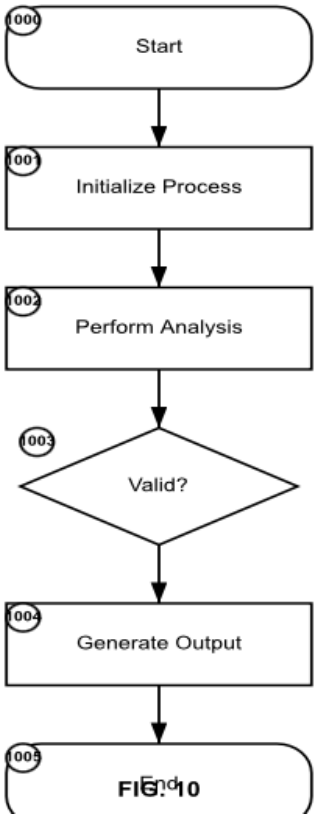

FIG. 10

FIG. 10 is a flowchart illustrating the intelligent patent claims generation according to an embodiment of the present invention.

FIG. 11 - Patent Strength Assessment Matrix

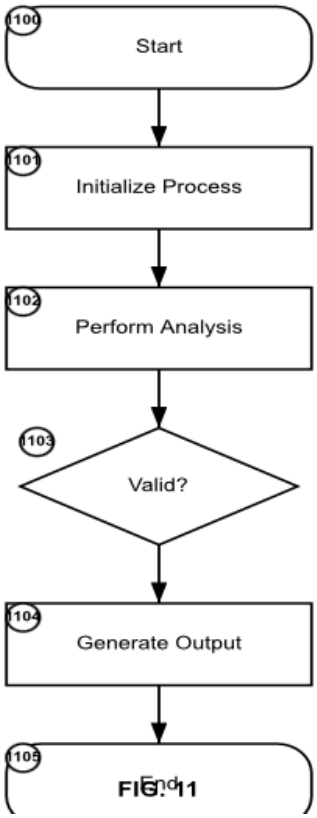

FIG. 11

FIG. 11 is a flowchart illustrating the patent strength assessment matrix according to an embodiment of the present invention.

FIG. 12 - Prior Art Differentiation Report Generator

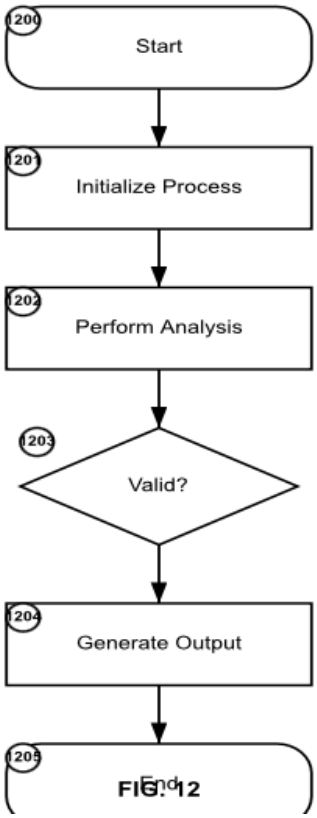

FIG. 12

FIG. 12 is a flowchart illustrating the prior art differentiation report generator according to an embodiment of the present invention.

FIG. 13 - Episode Progress Breakdown with Real-Time Aggregation

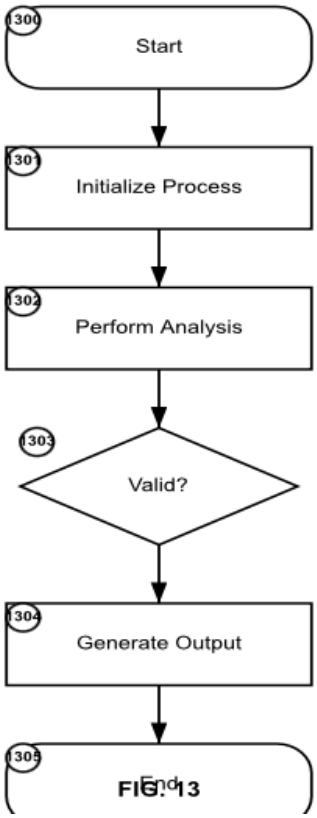

FIG. 13

FIG. 13 is a flowchart illustrating the episode progress breakdown with real-time aggregation according to an embodiment of the present invention.

FIG. 14 - Batch Recommendation System with Cost Optimization

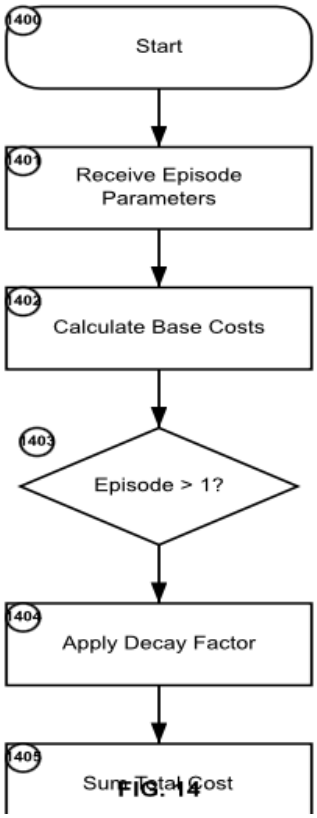

FIG. 14 is a flowchart illustrating the batch recommendation system with cost optimization according to an embodiment of the present invention.

FIG. 15 - Multi-Provider Integration Architecture

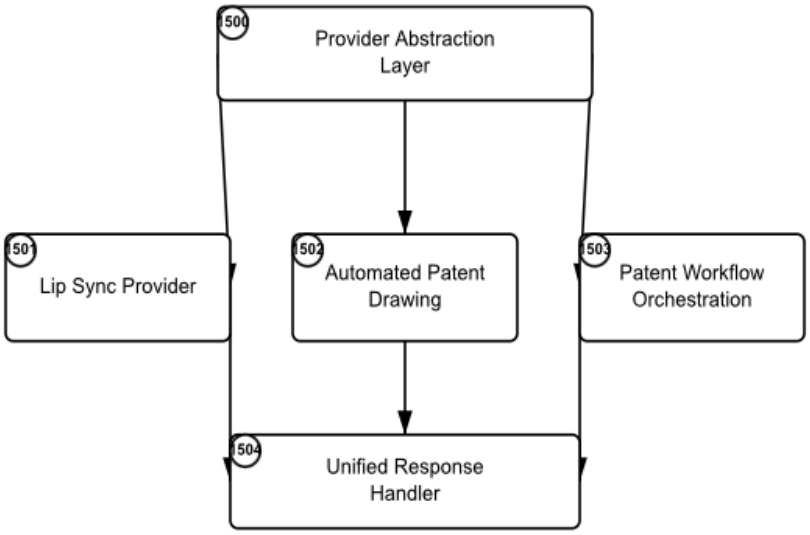

FIG. 15

FIG. 15 is a block diagram illustrating the multi-provider integration architecture with abstraction layers and service orchestration according to an embodiment of the present invention.

FIG. 16 - Data Structure Relationships

FIG. 16

FIG. 16 is an entity-relationship diagram illustrating the data structure relationships and multi-tenant isolation architecture according to an embodiment of the present invention.

FIG. 17 - Production Workflow Process

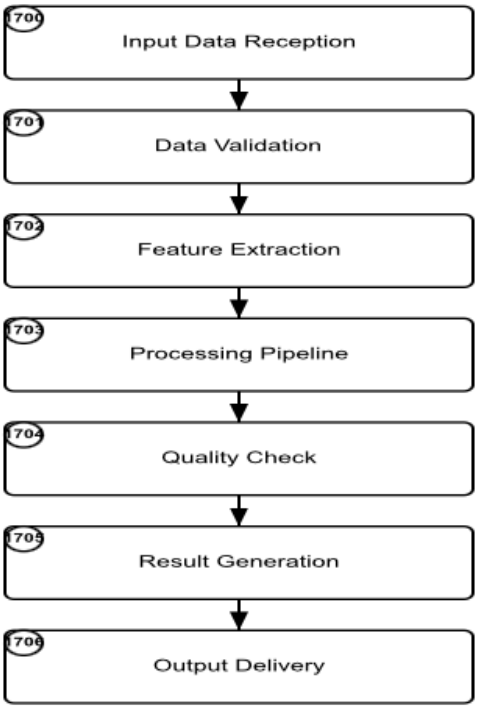

FIG. 17

FIG. 17 is a flowchart illustrating the production workflow process from input to final output according to an embodiment of the present invention.

FIG. 18 - User Interface Architecture

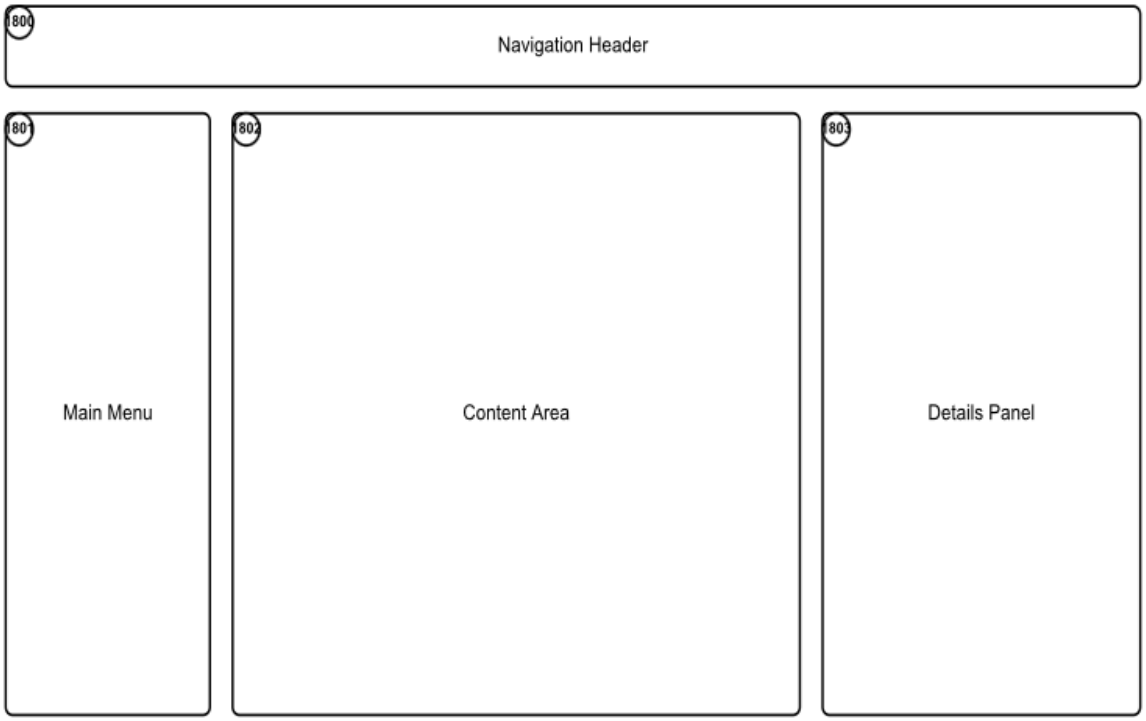

FIG. 18

FIG. 18 is a wireframe illustration of the user interface architecture showing main navigation and workflow management components according to an embodiment of the present invention.